FDS EXPERIMENT 18

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Experiment to understand KNN Algorithm for a given data set.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
```

```
data = {
    "Study_Hours": [1, 2, 2, 3, 4, 5, 6, 6, 7, 8],
    "Attendance": [50, 55, 60, 65, 70, 75, 80, 85, 90, 95],
    "Result": [
        "Fail", "Fail", "Fail", "Fail", "Pass",
        "Pass", "Pass", "Pass", "Pass", "Pass"
    ]
}

df = pd.DataFrame(data)

print(df)
```

```
   Study_Hours  Attendance Result
0            1          50   Fail
1            2          55   Fail
2            2          60   Fail
3            3          65   Fail
4            4          70   Pass
5            5          75   Pass
6            6          80   Pass
7            6          85   Pass
8            7          90   Pass
9            8          95   Pass
```

```
X = df[["Study_Hours", "Attendance"]]
y = df["Result"]
```

```
model = make_pipeline(
    StandardScaler(),
    KNeighborsClassifier(n_neighbors=3)
)

model.fit(X, y)

print("KNN model trained successfully!")
```

```
KNN model trained successfully!
```

```
new_student = pd.DataFrame({
    "Study_Hours": [4.5],
    "Attendance": [72]
})

prediction = model.predict(new_student)

print("Predicted Result:", prediction[0])
```

```
Predicted Result: Pass
```